1. Structure from Motion Consistency

Given:

- **Dataset A:** Accurate depth but requires high-quality images.
- **Dataset B:** Inconsistent depth but works on low-quality images.

Answer:

Dataset A provides more accurate and consistent depth information, but it depends on high-quality images. **Dataset B** is more tolerant of poor-quality images but produces inconsistent depth.

For real-world deployment, **Dataset A is preferable when high-quality cameras and good lighting are available**, because accurate depth is important for reliable 3D reconstruction. However, in low-quality or uncontrolled environments, Dataset B may be more practical because of its robustness.

Decision: Choose **Dataset A for accuracy-critical applications** and **Dataset B when image quality is unpredictable**.

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2. Stereo Vision vs Motion Estimation

Given:

- **Stereo Vision:** Accurate but requires two cameras.
- **Motion Estimation:** Less accurate but works with one camera.

Answer:

Stereo Vision provides better depth accuracy because it obtains depth from two different viewpoints. However, it requires **two synchronized cameras**, increasing hardware cost and complexity.

Motion Estimation uses a **single camera**, making it cheaper, simpler, and suitable for devices with hardware limitations. However, its depth estimation is generally less accurate.

Decision: Under hardware constraints, **Motion Estimation is preferable** because it requires only one camera and reduces system cost and complexity.

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3. Model Generalization vs Overfitting

Given:

Model	Training	Testing
A	98%	70%
B	90%	88%

Answer:

Model A has very high training accuracy but much lower testing accuracy. This indicates **overfitting**, meaning the model performs well on training data but poorly on unseen data.

Model B has slightly lower training accuracy but much higher testing accuracy. The small difference between training and testing performance indicates **better generalization**.

Decision: Model B is more reliable for deployment because it performs consistently on unseen data and has a lower risk of overfitting.

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4. YOLO Performance Across Scales

Given:

- Large objects → **95%**
- Medium objects → **85%**
- Small objects → **60%**

Answer:

YOLO performs very well for **large objects (95%)** and reasonably well for **medium objects (85%)**, but its performance decreases significantly for **small objects (60%)**.

This shows that YOLO alone may not be sufficient for applications requiring reliable detection across all object sizes, especially small objects.

Improvements:

- Use higher-resolution input images.
- Use multi-scale feature detection.
- Use feature pyramid networks.
- Improve the training dataset with more small-object examples.
- Apply data augmentation.

Decision: YOLO is suitable for large and medium objects, but **additional multi-scale improvements are required for reliable small-object detection**.

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5. Motion Model Selection for Animation

Given:

- **Model A:** Smooth but physically inaccurate.
- **Model B:** Realistic but computationally intensive.

Answer:

Model A produces smooth animation with lower computational requirements, making it suitable for **real-time applications such as games, VR, and interactive systems**.

Model B provides more realistic physical motion but requires more computational resources. Therefore, it is better suited for **cinematic animation, movies, and high-quality offline rendering**, where realism is more important than real-time performance.

Decision:

Application	Preferred Model	Reason
Real-time games	**Model A**	Fast and smooth
VR/AR	**Model A**	Low computational cost
Interactive animation	**Model A**	Real-time response
Movies/Cinematic	**Model B**	High physical realism
Offline rendering	**Model B**	Computation is less restrictive

Conclusion: Choose **Model A when speed and real-time performance are important**, and **Model B when physical accuracy and realism are the priority**.